Analog - 1.1.18

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Question

A Hilbert transformer is a :

1. Non − linear system
2. Non−causal system
3. Time−varying system
4. Low−pass system

(EC 2007)

Solution

Hilbert Transform is defined as:

Time Domain:

The output (Hilbert transform) of any input $x(t)$ is given by convolution:

$$y(t) = x(t) * h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} \, d\tau \tag{1}$$

where , impulse response $h(t) = \frac{1}{\pi t}$, $-\infty < t < \infty$

Solution

Frequency Domain:

$$H(j\omega) = \begin{cases} -j = e^{-j\frac{\pi}{2}}, & \omega > 0 \\ 0, & \omega = 0 \\ +j = e^{j\frac{\pi}{2}}, & \omega < 0 \end{cases} \tag{2}$$

here, Magnitude : $|H(j\omega)| = 1$ Phase : $\angle H(j\omega) = \begin{cases} -90^\circ, & \omega > 0 \\ +90^\circ, & \omega < 0 \end{cases}$

Solution

Let $x_1(t)$, $x_2(t)$ be any two signals and a, b be any scalars, then $y_1(t)$, $y_2(t)$ are the Hilbert transform outputs for the signals $x_1(t)$, $x_2(t)$ respectively:

$$y_1(t) = x_1(t) * h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x_1(\tau)}{t-\tau} \, d\tau \tag{3}$$

$$y_2(t) = x_2(t) * h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x_2(\tau)}{t-\tau} \, d\tau \tag{4}$$

Now, apply the Hilbert Transform to $ax_1(t) + bx_2(t)$ and let $z(t)$ be the output:

$$z(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{ax_1(\tau) + bx_2(\tau)}{t-\tau} \, d\tau \tag{5}$$

Solution

Using linearity of integration:

$$z(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{ax_1(\tau)}{t-\tau}\,d\tau + \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{bx_2(\tau)}{t-\tau}\,d\tau \tag{6}$$

$$= \frac{a}{\pi}\int_{-\infty}^{\infty} \frac{x_1(\tau)}{t-\tau}\,d\tau + \frac{b}{\pi}\int_{-\infty}^{\infty} \frac{x_2(\tau)}{t-\tau}\,d\tau \tag{7}$$

$$= ay_1(t) + by_2(t) \tag{8}$$

Therefore, both additivity and homogeneity are satisfied by Hilbert Transform. Thus, **Hilbert transform is linear**.

Solution

from (1) , for input $x(t)$

$$y(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau} \, d\tau \tag{9}$$

on shifting the output :

$$x_1(t) = x(t - t_0) \tag{10}$$

then ,

$$y_1(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau - t_0)}{t-\tau} \, d\tau \tag{11}$$

Let:

$$\lambda = \tau - t_0$$
$$\Rightarrow \quad \tau = \lambda + t_0, \quad d\tau = d\lambda$$

Solution

Substituting:

$$y_1(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t-(\lambda+t_0)}\,d\lambda \tag{12}$$

$$y_1(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{(t-t_0)-\lambda}\,d\lambda \tag{13}$$

On comparing with original output :

$$y(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t-\lambda}\,d\lambda \tag{14}$$

Thus,

$$y_1(t) = y(t-t_0) \tag{15}$$

Hence, the Hilbert Transform is Time-Invariant

Solution

The impulse response of the Hilbert transformer is:

$$h(t) = \frac{1}{\pi t} \tag{16}$$

For any $t < 0$, substitute a specific value, say $t = -1$:

$$h(-1) = \frac{1}{\pi \cdot (-1)} = -\frac{1}{\pi} \neq 0 \tag{17}$$

More generally, for *any* $t < 0$:

$$h(t) = \frac{1}{\pi t} \neq 0 \qquad \forall\, t < 0 \tag{18}$$

since $\frac{1}{\pi t}$ is nonzero for all $t \neq 0$.

$$\textbf{Required for causality:} h(t) = 0 \quad \forall\, t < 0 \tag{19}$$

$$\textbf{Actual impulse response:} h(t) = \frac{1}{\pi t} \neq 0 \quad \forall\, t < 0 \tag{20}$$

The condition is **not satisfied**.

Solution

The output of the Hilbert transformer at time t is:

$$y(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau}\, d\tau = \frac{1}{\pi}\int_{-\infty}^{t} \frac{x(\tau)}{t-\tau}\, d\tau + \underbrace{\frac{1}{\pi}\int_{t}^{\infty} \frac{x(\tau)}{t-\tau}\, d\tau}_{\text{depends on future input}}$$

The second integral involves $x(\tau)$ for $\tau > t$, i.e., future values of the input. This is a confirmation of non-causality.

Since $h(t) \neq 0$ for $t < 0$, the causality condition is violated:

The Hilbert transformer is a **Non-Causal** system.

Solution

For Low-pass system :

$$|H(\omega)| = 1 \quad \text{for } |\omega| < \omega_c \qquad \text{and} \qquad |H(\omega)| = 0 \quad \text{for } |\omega| > \omega_c \tag{21}$$

For High-pass :

$$|H(\omega)| = 0 \quad \text{for } |\omega| < \omega_c \qquad \text{and} \qquad |H(\omega)| = 1 \quad \text{for } |\omega| > \omega_c \tag{22}$$

For Band-pass :

$$|H(\omega)| = 1 \quad \text{for } \omega_1 < |\omega| < \omega_2 \qquad \text{and} \qquad |H(\omega)| = 0 \quad \text{elsewhere} \tag{23}$$

For All-pass :

$$|H(\omega)| = 1 \quad \text{for all } \omega \tag{24}$$

The frequency response is:

$$H(\omega) = -j \cdot \text{sgn}(\omega) \quad \text{from (2)} \tag{25}$$

Solution

Computing the magnitude response:

$$|H(\omega)| = |-j \cdot \text{sgn}(\omega)| = |-j| \cdot |\text{sgn}(\omega)| = 1 \cdot 1 = 1 \qquad \forall\, \omega \neq 0 \quad (26)$$

Since $|H(\omega)| = 1$ for all ω, the all-pass condition is satisfied directly.

The Hilbert transformer is an **All-pass** system .

Conclusion :
Thus , Hilbert transform is a linear time invariant , non-causal , all-pass system .
(option 2 is correct)

Python code

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import hilbert

# example taken for time domain
fs = 1000
T = 1.0
t = np.linspace(0, T, fs, endpoint=False)
x = np.cos(2 * np.pi * 10 * t)
analytic = hilbert(x)
y = np.imag(analytic)

# frequency domain
omega = np.linspace(-10, 10, 10000)
H_real = np.zeros_like(omega)
H_imag = -np.sign(omega)
H_mag = np.where(omega == 0, 0, 1)
```

Python Code

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# plot 1
axes[0, 0].plot(t, x, color='royalblue', linewidth=2,label=r'$x(t
    ) = \cos(2\pi \cdot 10t)$')
axes[0, 0].plot(t, y, color='crimson', linewidth=2, linestyle='--
    ',label=r'$y(t) = \sin(2\pi \cdot 10t)$')
axes[0, 0].set_xlim(0, 0.2)
axes[0, 0].set_xlabel('$t$ (s)', fontsize=12)
axes[0, 0].set_ylabel('Amplitude', fontsize=12)
axes[0, 0].set_title(r'Time Domain  $x(t)$ and $y(t)$',fontsize
    =12)
axes[0, 0].legend(fontsize=9)
axes[0, 0].grid(True, alpha=0.3)

# plot 2
axes[0, 1].plot(omega, H_real, color='darkorange', linewidth=2,
    label=r'$\mathrm{Re}[H(\omega)] = 0$')
```

Python

```
axes[0, 1].axhline(0, color='black', linewidth=0.6)
axes[0, 1].axvline(0, color='black', linewidth=0.5, linestyle='--
    ', alpha=0.4)
axes[0, 1].set_xlim(-10, 10)
axes[0, 1].set_ylim(-1.8, 1.8)
axes[0, 1].set_yticks([-1, 0, 1])
axes[0, 1].set_xlabel(r'$\omega$', fontsize=12)
axes[0, 1].set_ylabel(r'$\mathrm{Re}[H(\omega)]$', fontsize=12)
axes[0, 1].set_title(r'Frequency Domain  $\mathrm{Re}[H(\omega)]
    = 0$',fontsize=12)
axes[0, 1].legend(fontsize=9)
axes[0, 1].grid(True, alpha=0.3)

# plot 3
axes[1, 0].plot(omega, H_imag, color='crimson', linewidth=2,
               label=r'$\mathrm{Im}[H(\omega)] = -\mathrm{sgn}(\
                   omega)$')
axes[1, 0].plot(0, 0, 'ko', markersize=5)
axes[1, 0].axhline(0, color='black', linewidth=0.6)
```

Python code

```
axes[1, 0].axvline(0, color='black', linewidth=0.5,linestyle='--'
    , alpha=0.4)
axes[1, 0].set_xlim(-10, 10)
axes[1, 0].set_ylim(-1.8, 1.8)
axes[1, 0].set_yticks([-1, 0, 1])
axes[1, 0].set_xlabel(r'$\omega$', fontsize=12)
axes[1, 0].set_ylabel(r'$\mathrm{Im}[H(\omega)]$', fontsize=12)
axes[1, 0].set_title(r'Frequency Domain  $\mathrm{Im}[H(\omega)]
    = -\mathrm{sgn}(\omega)$',fontsize=12)
axes[1, 0].legend(fontsize=9)
axes[1, 0].grid(True, alpha=0.3)
axes[1, 0].annotate('+1', xy=(-9, 1.1), fontsize=10, color='
    crimson')
axes[1, 0].annotate('-1', xy=( 5, -1.2), fontsize=10, color='
    crimson')

# plot 4
axes[1, 1].plot(omega, H_mag, color='seagreen', linewidth=2.5,
    label=r'$|H(\omega)| = 1$')
```

Python code

```
axes[1, 1].plot(0, 0, 'ko', markersize=5, label=r'$H(0) = 0$')
axes[1, 1].axvline(0, color='black', linewidth=0.5,
                   linestyle='--', alpha=0.4)
axes[1, 1].fill_between(omega, H_mag, alpha=0.12, color='green')
axes[1, 1].set_xlim(-10, 10)
axes[1, 1].set_ylim(-0.3, 1.8)
axes[1, 1].set_yticks([0, 0.5, 1.0])
axes[1, 1].set_xlabel(r'$\omega$', fontsize=12)
axes[1, 1].set_ylabel(r'$|H(\omega)|$', fontsize=12)
axes[1, 1].set_title(r'Magnitude  $|H(\omega)| = 1$ (All-pass)',
                     fontsize=12)
axes[1, 1].legend(fontsize=10)
axes[1, 1].grid(True, alpha=0.3)


plt.tight_layout()
plt.show()
```

Plot

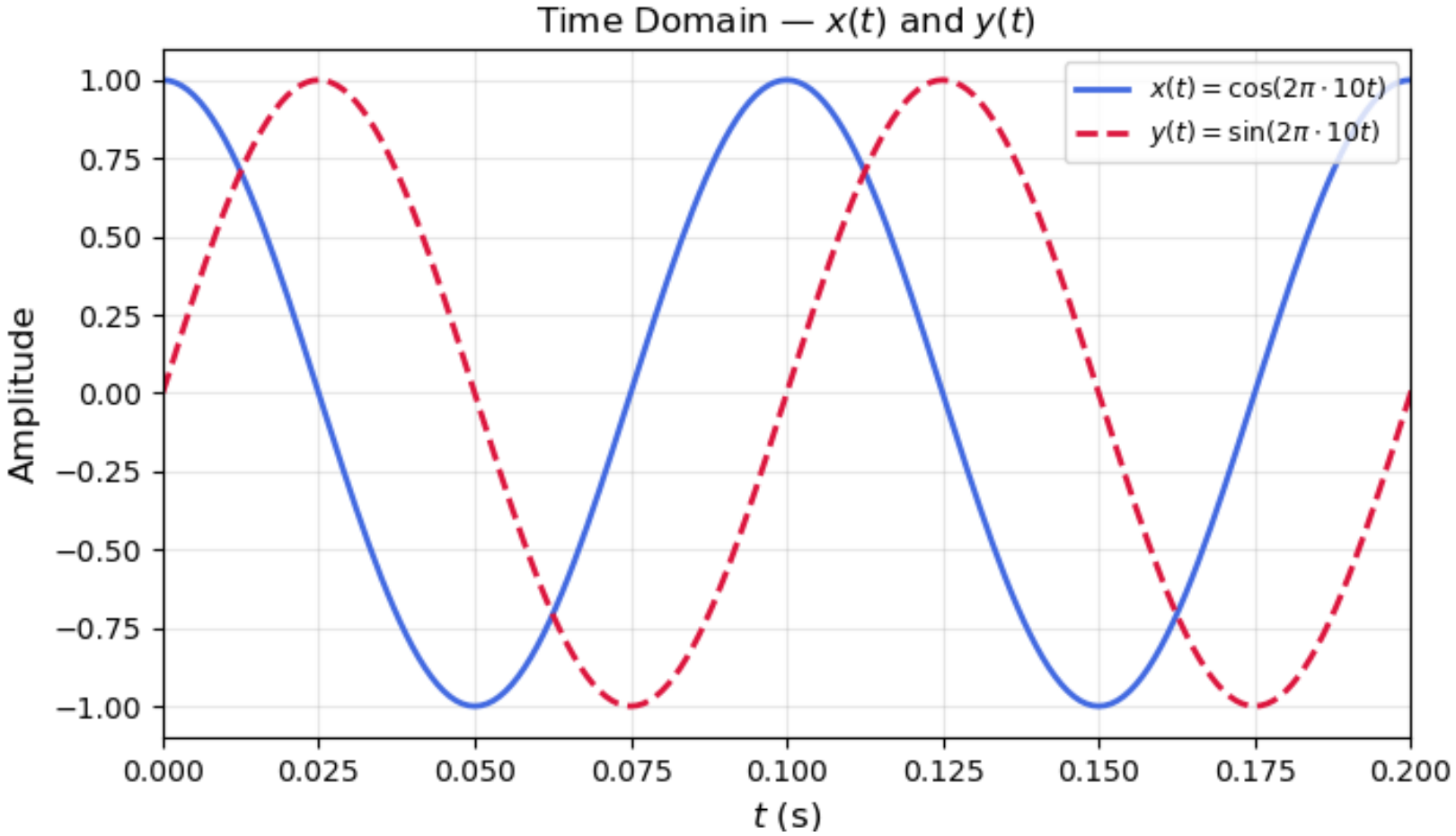
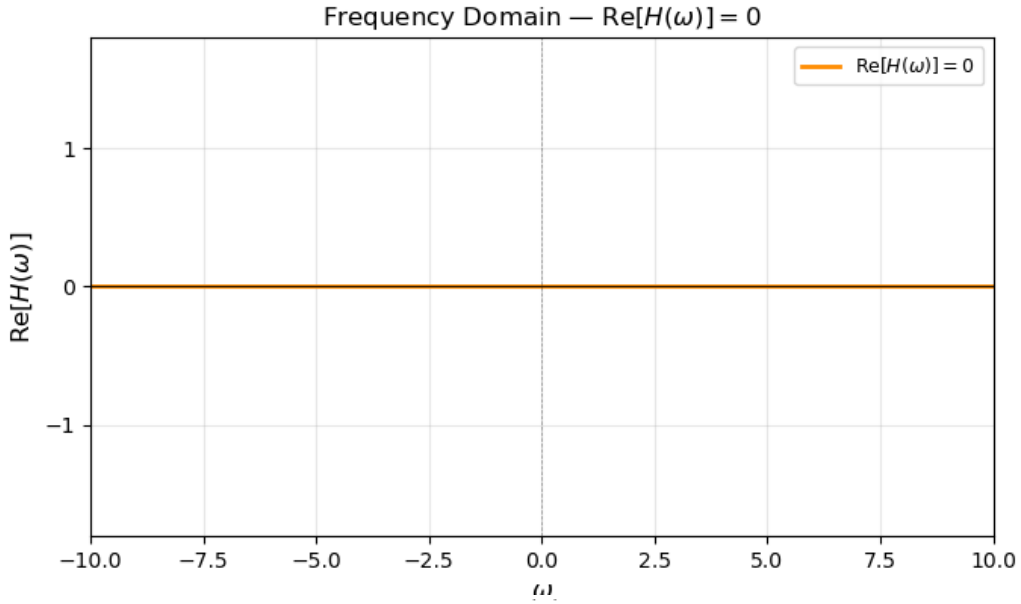
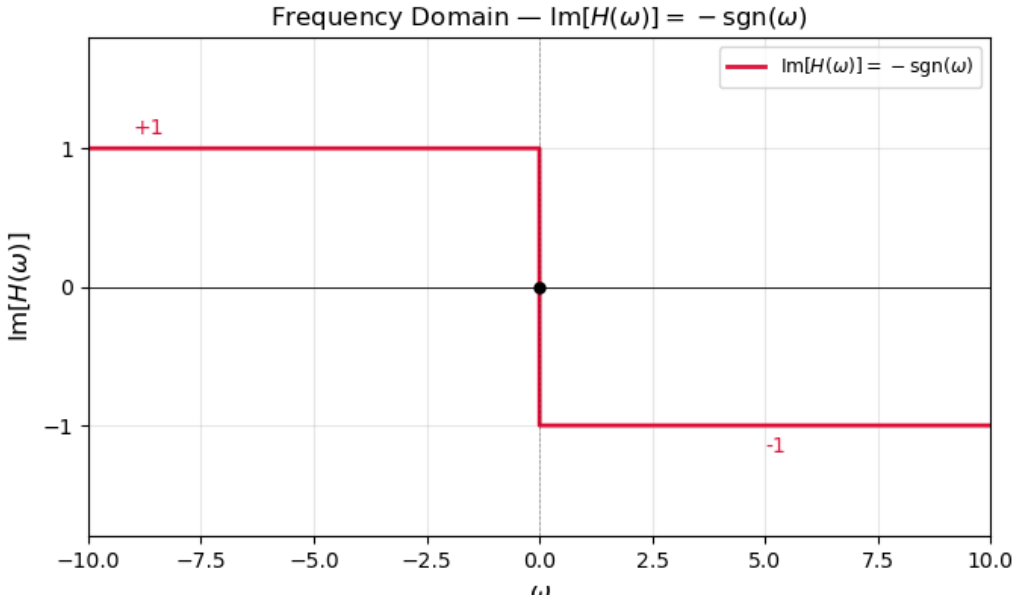


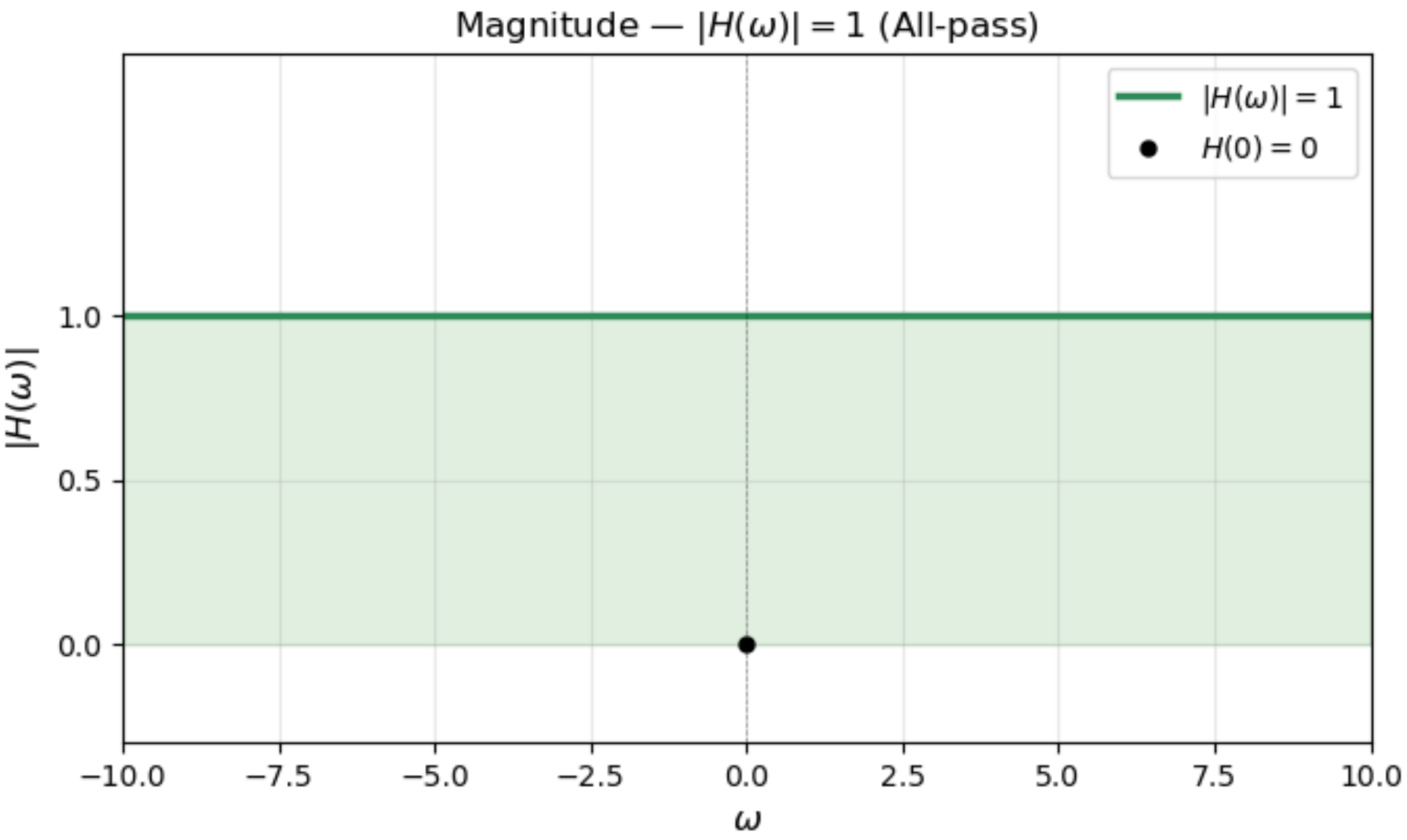
Figure: Example taken $x(t) = \cos(2\pi \cdot 10t)$

Plot


Frequency Domain — Re[$H(\omega)$] = 0
Re[$H(\omega)$] = 0
Re[$H(\omega)$]
1
0
−1
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0


Frequency Domain — Im[$H(\omega)$] = − sgn(ω)
Im[$H(\omega)$] = − sgn(ω)
+1
-1
Im[$H(\omega)$]
1
0
−1
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

Plot


Magnitude — $|H(\omega)| = 1$ (All-pass)
$|H(\omega)| = 1$
$H(0) = 0$
1.0
0.5
0.0
$|H(\omega)|$
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
ω